Inflation-Aware Portfolio Optimization

Balancing Downside Risk and Long-Term Purchasing Power

A Walk-Forward Study of Canadian-Listed ETFs Using CVaR and Return Constraints

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Abstract

This study examines the trade-off between downside-risk control and inflation-adjusted purchasing-power growth in portfolios of Canadian-listed exchange-traded funds (ETFs). It compares equal weighting, global minimum variance (GMV), minimum Conditional Value-at-Risk (CVaR), and CVaR strategies subject to minimum estimated-return constraints. Seven ETFs cover Canadian, U.S., developed international and emerging-market equities, Canadian bonds, and gold. Monthly returns span May 2013–September 2026, and 60-month rolling training windows produce 101 monthly historical test observations from May 2018–September 2026. Canadian CPI permits 99 matched months of real-return evaluation (June 2018–August 2026).

Unconstrained expanded minimum-CVaR achieved 4.08% annualized nominal return with 1.56% historical monthly CVaR (95%), while the 8% return-target strategy achieved 9.63% return and 3.38% CVaR. Expanded equal weighting achieved 9.72% return and 4.41% CVaR. The 8% strategy had higher observed one- and three-year inflation-beating frequencies than equal weighting, but a retrospective exposure-matched static benchmark had lower volatility, CVaR and maximum drawdown. Trading-cost and gold-weight-cap sensitivity experiments produced limited changes in the central historical patterns. These are exploratory, overlapping-window historical results, not forecasts or a demonstration of universal portfolio optimality.

Keywords: Portfolio optimization; CVaR; inflation; purchasing power; operations research; walk-forward backtesting; ETF allocation

1. Introduction

A portfolio with low nominal volatility can still lose purchasing power when returns fail to keep pace with inflation. Conversely, a portfolio that grows real wealth may suffer losses unacceptable to investors with near-term liquidity needs or limited tolerance for drawdowns. Traditional mean–variance optimization emphasizes overall return variability, while CVaR emphasizes the unfavorable tail. Neither unconstrained risk-minimization objective directly guarantees real-wealth growth.

The central research question is whether requiring a minimum historical estimated return within a CVaR framework offers a more attractive balance between downside risk and inflation-adjusted investment outcomes. A secondary question is whether dynamic optimization adds value beyond simpler portfolios with similar asset-class exposures.

1.1 Research Questions

RQ1. Do GMV and unconstrained minimum-CVaR portfolios preserve long-term purchasing power?

RQ2. How do increasing expected-return constraints affect realized return, CVaR, maximum drawdown, and historical inflation-beating frequency?

RQ3. How does dynamic constrained-CVaR compare with equal weighting and exposure-matched static allocation?

RQ4. How sensitive are results to transaction costs, gold-weight limits, and selected stress periods?

2. Data and Experimental Design

2.1 Asset universe

ETF	Asset exposure	Universe
XIC.TO	Canadian equities	Core + Expanded
VFV.TO	U.S. S&P 500 equities	Core + Expanded
XEF.TO	Developed equities outside North America	Core + Expanded

XEC.TO	Emerging-market equities	Core + Expanded
XBB.TO	Canadian aggregate bonds	Core + Expanded
CGL-C.TO	Unhedged gold	Expanded
XSB.TO	Canadian short-duration bonds	Expanded

Adjusted ETF prices were retrieved using yfinance/Yahoo Finance and resampled to calendar month-end. They approximate distribution-adjusted investment returns; the analysis does not represent after-tax realized investor returns. The reported calculations use a saved processed return dataset.

2.2 Study timeline

Dataset / evaluation	Period	Months
ETF monthly returns	May 2013–September 2026	161
Rolling estimation window	Trailing months before each test	60
Nominal walk-forward returns	May 2018–September 2026	101
CPI index	May 2018–August 2026	100
Matched inflation observations	June 2018–August 2026	99

Inflation is measured using Statistics Canada, Table 18-10-0004-01, Canada All-items CPI, monthly not seasonally adjusted (2002=100). CPI is used ex post to evaluate purchasing power; it does not enter portfolio optimization as a trading signal. The matched CPI interval reflects an index change of 27.2864%.

3. Mathematical Methodology

3.1 Basic assumptions

For N assets, w denotes portfolio weights and r_t denotes the vector of monthly ETF returns. Portfolio return is $R_{p,t} = w^T r_t$, subject to $\Sigma_i w_i = 1$ and $w_i \geq 0$. Portfolios are unlevered, long-only, and rebalanced monthly to target weights. Equal-weight benchmarks use 1/N per ETF.

3.2 Global Minimum Variance

The GMV model solves $\min_{(w)} w^T \Sigma w$ subject to the budget and long-only constraints, where Σ is estimated from the preceding 60 monthly observations. Variance treats upside and downside deviations symmetrically.

3.3 Minimum Conditional Value-at-Risk

Let $L_t = -r_t^T w$ denote scenario loss, α = 0.95, η an auxiliary loss threshold, and $u_t \geq 0$ scenario excess-loss variables. The Rockafellar–Uryasev linear program minimizes $\eta + [1/((1-\alpha)T)]\Sigma_t u_t$ subject to $u_t \geq L_t - \eta$, $\Sigma_i w_i = 1$ and $w_i \geq 0$. A 60-month estimation window and a 95% tail level place strong influence on approximately three unfavorable months.

3.4 Return-constrained CVaR

The constrained version adds $\mu^T w \geq (1+a)^{(1/12)} - 1$, where μ îs the training-period arithmetic mean of monthly returns and a is the nominal annual target (0%, 2%, 4%, 6%, or 8%). This is a constraint on an estimated monthly arithmetic mean using a geometrically converted annual threshold: it is not a guarantee of realized annual compounded return or CPI outperformance. The model was solved with CVXPY and CLARABEL.

3.5 Chronological walk-forward evaluation

Each test month uses the 60 prior months to estimate moments and optimize portfolio weights. The realized return is then calculated using only the following test month. This is repeated 101 times. Audit tests verified that each training window excludes its test observation. However, retrospective strategy selection across the full historical evaluation still presents model-selection bias.

3.6 Performance definitions

Geometric annual return: $[\Pi_t(1+R_t)]^{(12/T)} - 1$. Annualized volatility: standard deviation of monthly returns × $\sqrt{12}$. Maximum drawdown: greatest decrease in compounded wealth from a preceding peak including initial wealth = 1. Historical monthly CVaR 95%: average of empirical worst-tail monthly losses, with fractional weighting as needed. Real monthly return: $(1+R_t)/(1+\pi_t) - 1$. Inflation-beating frequency: share of overlapping H-month rolling windows with positive compounded real return, for H = 12, 36, 60.

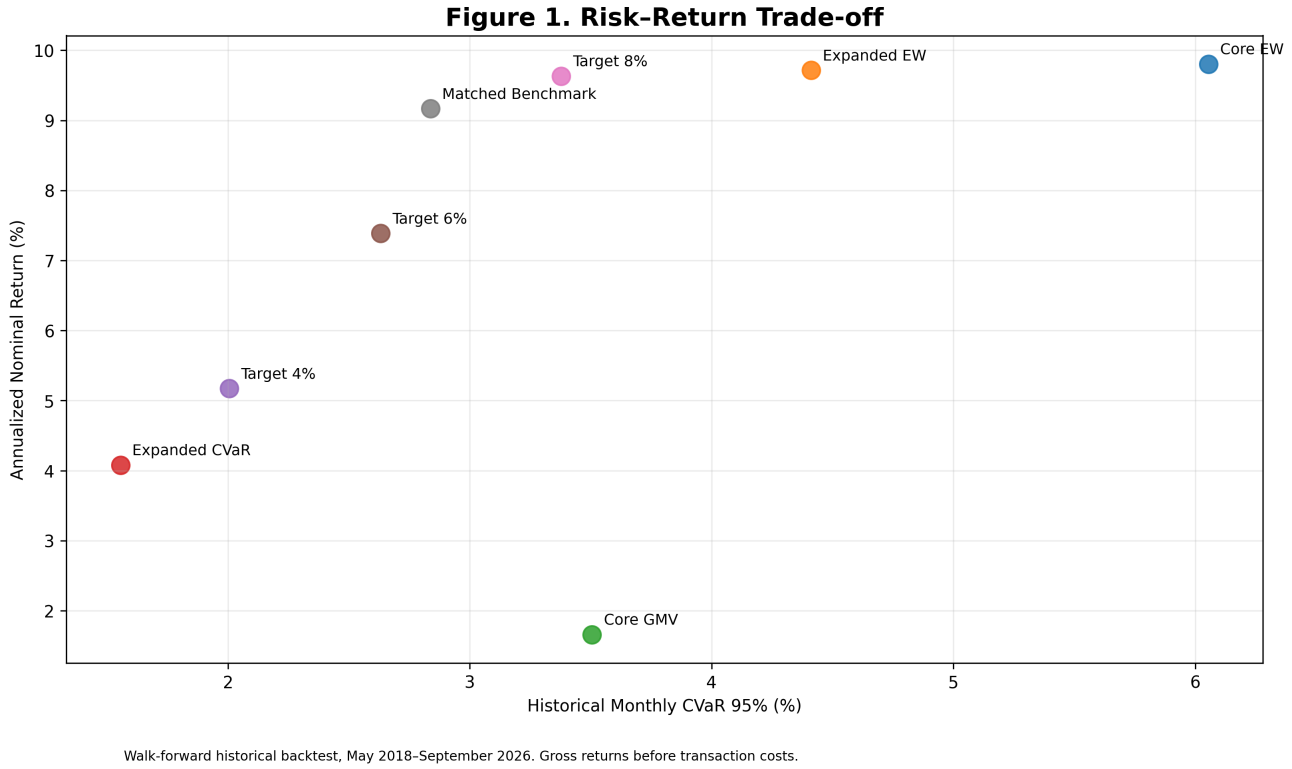
4. Results

4.1 Risk–return trade-offs

Portfolio	Annual return %	Volatility %	Max DD %	CVaR 95%*
Core EW	9.80	10.21	16.42	6.06
Expanded EW	9.72	7.92	12.65	4.41
Core GMV	1.66	6.03	15.25	3.51
Expanded GMV	2.24	2.24	6.08	1.32
Core CVaR	2.78	6.25	14.13	3.65
Expanded CVaR	4.08	3.06	6.11	1.56
Target 0%	4.08	3.06	6.11	1.56
Target 2%	4.03	3.13	6.47	1.55
Target 4%	5.17	3.96	7.09	2.01
Target 6%	7.39	5.37	8.01	2.63
Target 8%	9.63	7.00	9.37	3.38
Matched benchmark†	9.17	6.30	8.05	2.84

* Historical monthly CVaR, not annualized. † Retrospectively constructed diagnostic comparison. Other statistics cover 101 nominal test months.

Figure 1. Risk–Return Trade-off

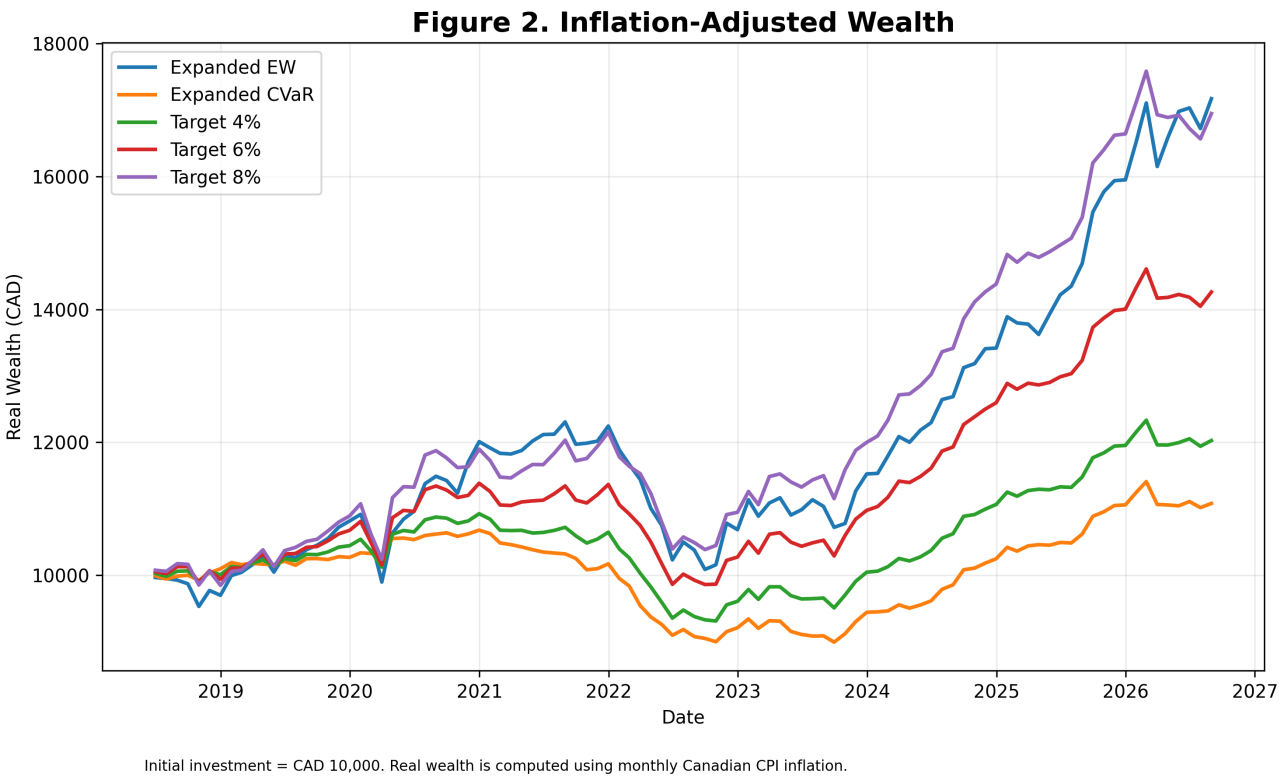


Minimum-risk portfolios achieved relatively low nominal tail losses but often substantially reduced returns. The Target 8% portfolio earned 9.63% annually, close to Expanded EW's 9.72%, with lower CVaR (3.38% versus 4.41%). Nevertheless, a static exposure-matched benchmark earned 9.17% with even lower CVaR (2.84%), showing that dynamic optimization does not dominate simple allocation on all criteria.

4.2 Purchasing power

Portfolio	Real annual return %
Core EW	6.73
Expanded EW	6.77
Core GMV	−1.23
Expanded GMV	−0.59
Core CVaR	−0.14
Expanded CVaR	1.25
Target 4%	2.26
Target 6%	4.40
Target 8%	6.60
Matched benchmark	6.36

Figure 2. Inflation-Adjusted Wealth



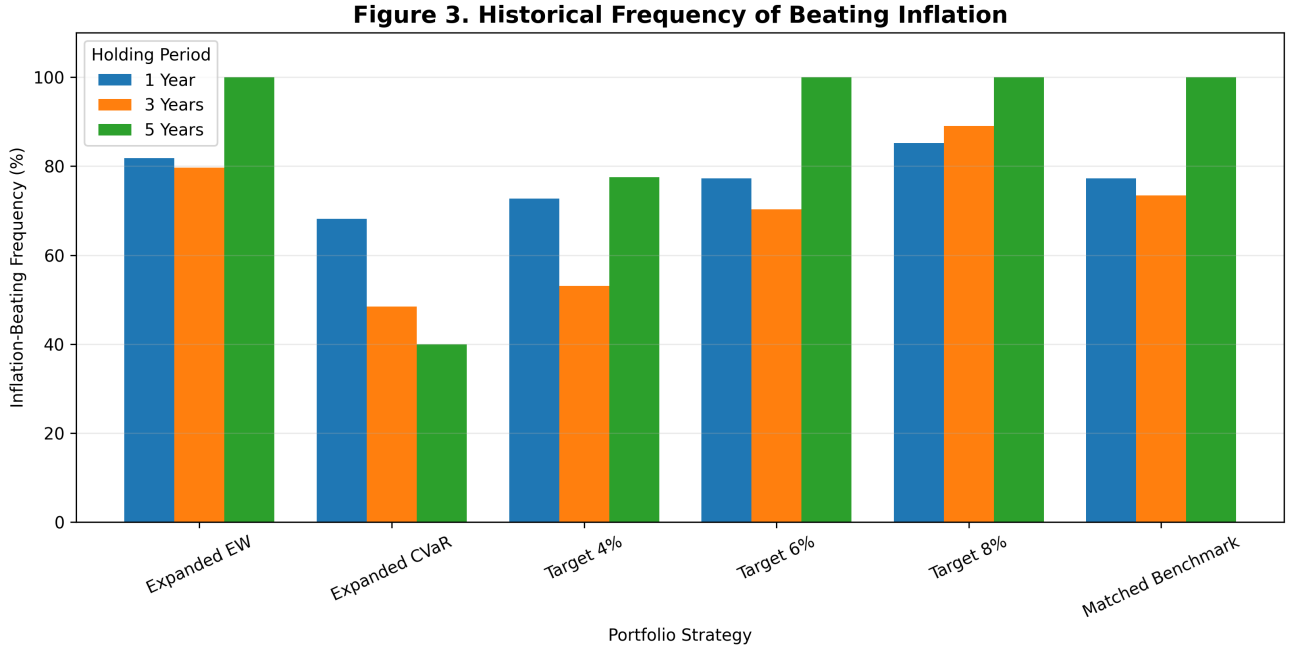
Initial investment = CAD 10,000. Real wealth is computed using monthly Canadian CPI inflation.

Unconstrained nominal risk minimization did not consistently produce strong real-wealth growth. Target 8% and Expanded EW accumulated substantially more purchasing power than pure minimum-CVaR but experienced more pronounced interim fluctuations. Real-return results use 99 matched months, not the 101-month nominal evaluation.

4.3 Historical frequency of beating CPI

Portfolio	1-year %	3-year %	5-year %
Expanded EW	81.82	79.69	100.00
Expanded CVaR	68.18	48.44	40.00
Target 4%	72.73	53.12	77.50
Target 6%	77.27	70.31	100.00
Target 8%	85.23	89.06	100.00
Matched benchmark	77.27	73.44	100.00

Figure 3. Historical Frequency of Beating Inflation



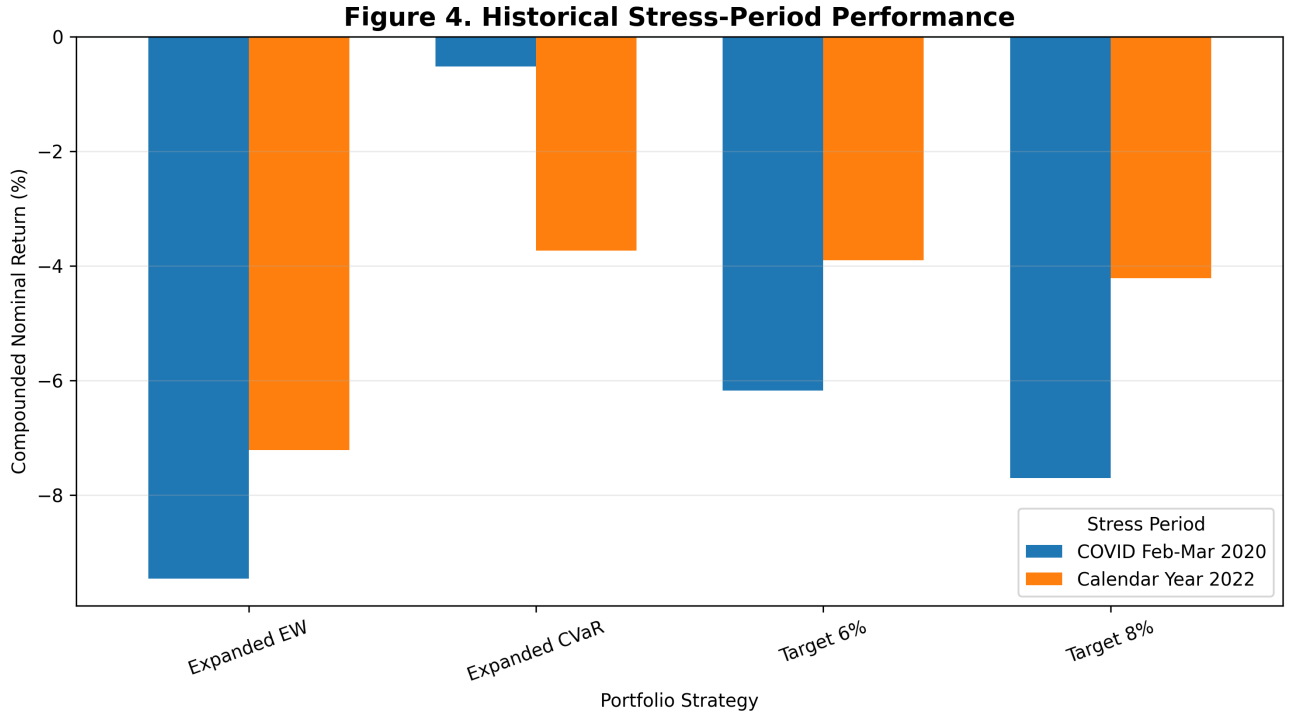
Rolling real returns, June 2018–August 2026. Overlapping windows; historical frequencies, not forecasts.

Among displayed portfolios, Target 8% had the highest observed one- and three-year positive-real-return frequencies. These windows are heavily overlapping, especially for five years, and must not be interpreted as independent future success probabilities. All 40 five-year windows were positive for multiple strategies.

4.4 Stress periods

Portfolio	Feb–Mar 2020 %	Calendar 2022 %
Expanded EW	−9.46	−7.22
Expanded CVaR	−0.52	−3.73
Target 6%	−6.18	−3.90
Target 8%	−7.70	−4.21

Figure 4. Historical Stress-Period Performance



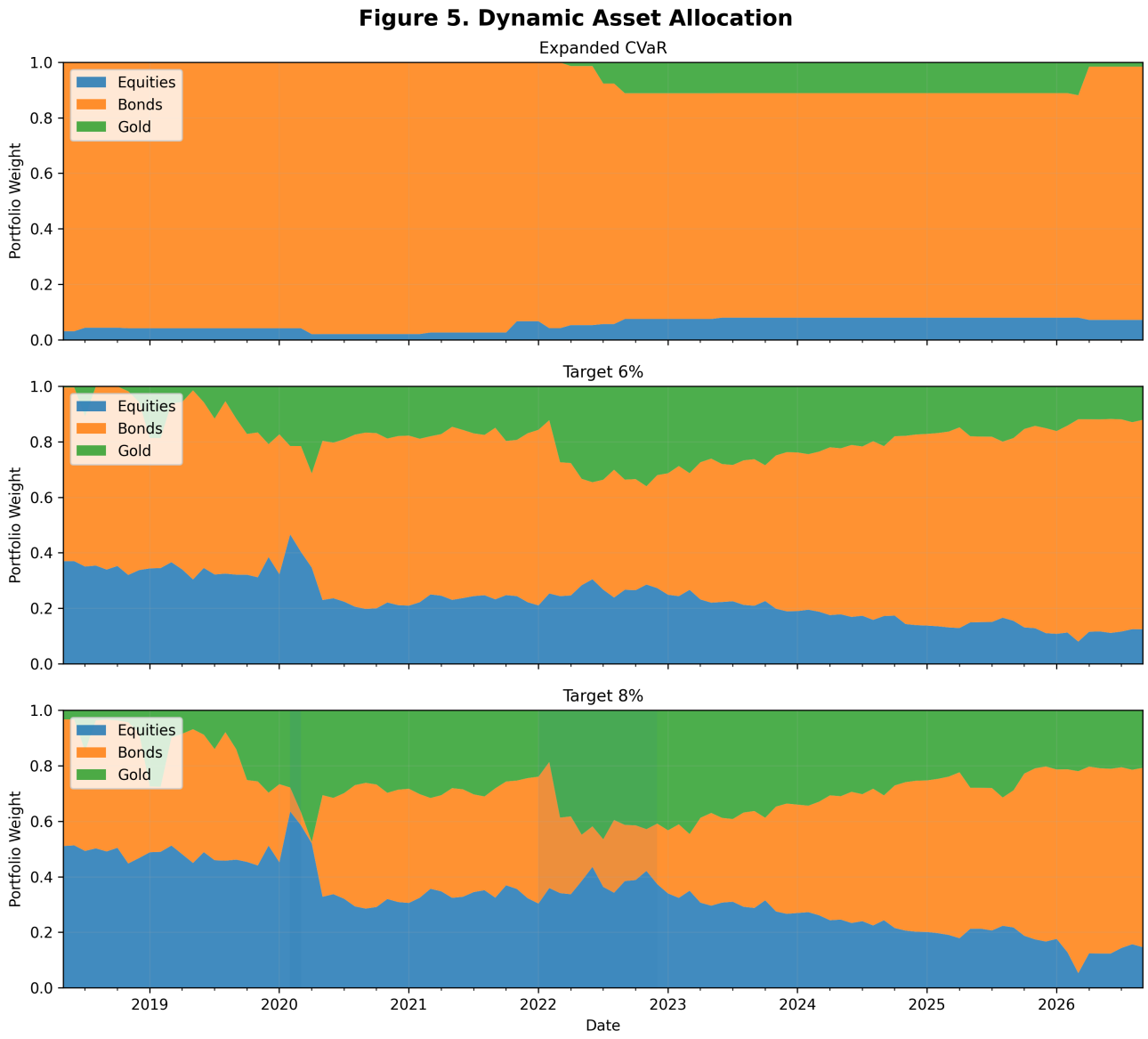
Monthly compounded nominal returns. COVID window: Feb-Mar 2020; 2022 window: Jan-Dec 2022. Before transaction costs.

Expanded CVaR experienced the smallest loss in both selected stress periods. Target 8% experienced smaller losses than Expanded EW in both episodes, though greater losses than Target 6%. These selected periods overlap the overall evaluation and provide diagnostic evidence, not independent validation.

4.5 Allocation dynamics

Portfolio	Equities %	Bonds %	Gold %
Expanded CVaR	5.75	89.25	4.99
Target 6%	23.31	58.37	18.32
Target 8%	32.60	40.38	27.02

Figure 5. Dynamic Asset Allocation



Increased return constraints shifted allocations from bonds toward equities and gold. Target 8% allocated an average of 27.02% to gold (median 27.94%); 39 of 101 target months had gold weights ≥30%, with a maximum of 47.40%. These are descriptive historical allocation patterns, not evidence that the model predicted specific crises.

5. Implementation and Robustness

5.1 Model feasibility and consistency

All five original return targets were feasible in 101/101 training windows, and neither 30% nor 40% gold cap triggered fallback. The Target 0% and unconstrained Expanded CVaR monthly returns differed by at most 1.66×10^{-8}, indicating close numerical agreement. Feasibility concerns estimated in-sample returns, not realized future returns.

5.2 Trading costs

Portfolio	Avg monthly turnover %	Annual turnover	0 bps CAGR %	10 bps CAGR %	20 bps CAGR %
Expanded EW	0.966	0.116	9.715	9.702	9.690
Expanded CVaR	0.673	0.081	4.080	4.072	4.063
Target 6%	6.535	0.784	7.387	7.303	7.219
Target 8%	7.565	0.908	9.631	9.532	9.433

Turnover compares target allocations with pre-trade allocations after previous-month return drift. Hypothetical 0, 5, 10 and 20 basis-point costs are applied per unit of one-way turnover; initial establishment costs and investor-specific taxes are not included. The higher-turnover Target 8% strategy loses about 0.20 percentage points of annualized return under the 20 bps assumption, without overturning the broad observed trade-offs.

5.3 Gold caps

Gold limit	Return %	Volatility %	Max DD %	CVaR %	Feasible months
Uncapped Target 8%	9.63	7.00	9.37	3.38	101
30%	9.61	7.06	9.32	3.52	101
40%	9.69	7.01	9.03	3.39	101

Adding gold upper bounds of 30% or 40% produced relatively small changes in nominal risk-return statistics. These scenarios do not establish an optimal cap; they are retrospective sensitivity checks.

5.4 Static exposure-matched benchmark

A static benchmark used 32.60% equities (equal amounts in XIC, VFV, XEF, XEC), 40.38% XSB bonds, and 27.02% CGL-C gold. This allocation was derived from Target 8% average historical weights, making it retrospective. It generated slightly lower real growth (6.36% vs 6.60% for Target 8%) and lower one-/three-year historical inflation-beating frequencies, but also lower volatility, drawdown and CVaR. The results do not isolate the incremental causal effect of CVaR optimization from asset selection and dynamic allocation.

6. Discussion and Limitations

The findings distinguish nominal risk minimization from real purchasing-power preservation. Return requirements may correct excessive defensiveness but expose investors to greater monthly tail losses. Low risk alone is not sufficient for real growth, and complexity alone does not ensure superior outcomes relative to simple asset allocation.

Key limitations include exploratory selection of ETFs and highlighted target levels after observing historical results; sensitive arithmetic mean return estimates; CVaR estimation from approximately three lower-tail months per 60-month training period; highly overlapping rolling real-return windows; idealized monthly execution; unified hypothetical trading costs; and no investor-specific taxation or liquidity constraints. Adjusted market prices can be revised. A reconciliation check of the frozen monthly price and return datasets found a maximum absolute difference of 4.45×10^{-16}, consistent with floating-point rounding. An earlier discrepancy observed during development could not be reproduced using the frozen files. Historical market data may still be revised by the data provider. The benchmark uses ex post average weights and is not an untouched investable comparator. No formal confidence intervals or out-of-sample significance tests are claimed.

CPI is applied solely as retrospective evaluation and is not a real-time trading signal. CPI measurement uses headline Canadian inflation and may differ from an individual investor's consumption basket. The backtest's chronological training/test separation was audited, but full independence from exploration is not demonstrated.

7. Conclusion

A 60-month rolling evaluation of Canadian-listed ETF portfolios reveals a substantive tension between minimizing nominal tail losses and protecting long-term real wealth. Unconstrained minimum-CVaR can provide strong downside protection but at the cost of limited purchasing-power growth. Return-constrained CVaR produces more growth-oriented portfolio configurations and, in this sample, the 8% target preserves much of equal weighting's nominal return while reducing historical monthly CVaR. However, a static exposure-matched benchmark achieved even lower downside-risk statistics, underscoring that optimization complexity does not automatically yield universally superior portfolios.

The main contribution is a transparent empirical framework comparing tail losses, real wealth, historical inflation-beating frequencies, stress performance, turnover, and allocation concentration. The appropriate interpretation is a historical risk-return trade-off, not an assurance of future inflation outperformance. Future work could use a prospectively fixed holdout protocol, alternative return estimators and CVaR windows, and explicit inflation-sensitive objectives.

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Yahoo Finance. Historical pricing and adjusted price information. https://finance.yahoo.com/

Appendix A. Reproducibility and Release Checklist

• Notebook 1: 01_data_collection.ipynb — ETF and CPI sourcing/processing.

• Notebook 2: 02_portfolio_optimization.ipynb — optimization, validation, analysis, figures.

• Processed inputs: core_returns.csv; expanded_returns.csv; canada_monthly_inflation.csv.

• Principal outputs: all_11_strategy_monthly_returns.csv; all_11_strategy_research_summary.csv.

• Five verified PNG images: figure1_risk_return_tradeoff.png, figure2_inflation_adjusted_wealth.png, figure3_inflation_beating_frequency.png, figure4_historical_stress_testing.png, figure5_asset_allocation_dynamics.png.

Reproducibility and Data Limitations

The five research figures are included in this report. The numerical results are based on frozen processed datasets and the documented Python workflow. The saved ETF prices and returns passed reconciliation checks, with a maximum absolute difference of approximately 4.45×10^{-16}. The data validation notebook passed independent fresh-kernel execution. Full optimization reproducibility and clean-environment dependency verification remain pending.